

Ahlfors 1

Vincent Wang

September 2025

1 Fundamentals

Find conditions on when the equation $az + b\bar{z} + c = 0$ has exactly one complex solution.

The idea is to conjugate the equation and treat the equations as a systems in $z, \bar{z}$. We have $\bar{b}z + \bar{a}\bar{z} + \bar{c} = 0$. Solving gives

$$\begin{bmatrix} z \\ \bar{z} \end{bmatrix} = \frac{1}{a\bar{a} - b\bar{b}} \begin{bmatrix} \bar{a} & -b \\ -\bar{b} & a \end{bmatrix} \begin{bmatrix} -c \\ -\bar{c} \end{bmatrix},$$

from which we have the condition $|a| \neq |b|$ and $z = \frac{-\bar{a}c + b\bar{c}}{a\bar{a} - b\bar{b}}$.

For $|a|, |b| < 1$ prove that $|\frac{a-b}{1-\bar{a}\bar{b}}| < 1$.

We have $(1 - |a|^2)(1 - |b|^2) > 0 \rightarrow 1 + |a|^2|b|^2 > |a|^2 + |b|^2 \rightarrow 1 - a\bar{b} - \bar{a}b + a\bar{a}b\bar{b} > a\bar{a} + b\bar{b} - a\bar{b} - \bar{a}b \rightarrow |1 - a\bar{b}|^2 > |a - b|^2 \rightarrow |1 - a\bar{b}| > |a - b|$.

Show that $|z - a| + |z + a| = 2|c|$ has a solution iff $|a| \leq |c|$ and find the minimum and maximum values of $|z|$ when such a solution exists.

(1) Suppose the equation has a solution. Then $2|a| \leq |a - z| + |a + z| = 2|c| \rightarrow |a| \leq |c|$.

(2) Suppose $|a| \leq |c|$. If $a = 0$, then $|z| = |c|$, for which the minimum and maximum values of $|z|$ are $|c|$. Else, $2|z| \leq |z - a| + |z + a| = 2|c| \rightarrow |z| \leq |c|$. This is achievable by $z = \frac{|c|}{|a|}a$: $|z - a| + |z + a| = |a| \cdot \left| \frac{|c|}{|a|} - 1 \right| + |a| \cdot \left| \frac{|c|}{|a|} + 1 \right| = 2|c|$.

Note $|z - a|^2 + |z + a|^2 = 2(|z|^2 + |a|^2)$, so $2(|z|^2 + |a|^2) \geq \frac{1}{2}(|z - a| + |z + a|)^2 = 2|c|^2$, so $|z|^2 \geq |c|^2 - |a|^2$, so $|z| \geq \sqrt{|c|^2 - |a|^2}$. This is achievable by $z = \frac{ai}{|a|}\sqrt{|c|^2 - |a|^2}$. $|z - a| + |z + a| = |a| \cdot \left| \frac{i}{|a|}\sqrt{|c|^2 - |a|^2} - 1 \right| + |a| \cdot \left| \frac{i}{|a|}\sqrt{|c|^2 - |a|^2} + 1 \right| = 2|c|$.

We can combine cases, so the minimum and maximum values of $|z|$ are $\sqrt{|c|^2 - |a|^2}, |c|$,

respectively.

2 Geometric Representation

The complex numbers a_1, a_2, a_3 form an equilateral triangle iff $a_1^2 + a_2^2 + a_3^2 = a_1a_2 + a_2a_3 + a_3a_1$.

(1) Suppose the equation holds. Let $b_i = a_{i+1} - a_i$ with indices taken mod 3. We have $b_1^2 + b_2^2 + b_3^2 = 0$ and $b_1 + b_2 + b_3 = 0$. If $b_1 = 0$, then $b_2^2 + b_3^2 = 0, b_2 + b_3 = 0 \rightarrow b_2, b_3 = 0$. If $b_1 = b_2 = b_3 = 0$, then the a_i 's are equal and form an equilateral triangle.

Else, assume $b_1 \neq 0$. Substitute for b_3 , so $b_1^2 + b_2^2 + (-b_1 - b_2)^2 = 0 \rightarrow 2b_1^2 + 2b_1b_2 + 2b_2^2 = 0 \rightarrow (b_2/b_1)^2 + (b_2/b_1) + 1 = 0$. Let ω be a root of this quadratic in b_2/b_1 , so $b_2 = \omega b_1$ and $b_3 = -(b_1 + b_2) = (-1 - \omega)b_1 = \omega^2 b_1$. Since $\omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1) = 0$, we have $|\omega| = 1$, so $|b_1| = |b_2| = |b_3|$. That is, the pairwise distances between a_1, a_2, a_3 are equal.

(2) Conversely, suppose $|b_1| = |b_2| = |b_3| = s$. If $s = 0$, we're done, so assume $s > 0$. Conjugate $b_1 + b_2 + b_3 = 0$ to get $s^2/b_1 + s^2/b_2 + s^2/b_3 = 0 \rightarrow b_1b_2 + b_2b_3 + b_3b_1 = 0$. Then $b_1^2 + b_2^2 + b_3^2 = (b_1 + b_2 + b_3)^2 - 2(b_1b_2 + b_2b_3 + b_3b_1) = 0$, so $a_1^2 + a_2^2 + a_3^2 = a_1a_2 + a_2a_3 + a_3a_1$.

Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is cont diff. Then $\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f((x+h)+iy) - f(x+iy)}{h} = f'(z)$.
 $\frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x+(y+h)i) - f(x+yi)}{h} = if'(z)$.

3 Partial Fractions

Claim: Let Q be a separable polynomial with roots α_i for $i \in [n]$. Let P be a polynomial of degree m less than n . Then

$$\frac{P(z)}{Q(z)} = \sum_{i \in [n]} \frac{P(\alpha_i)}{Q'(\alpha_i)(z - \alpha_i)}.$$

Proof: We have for $\zeta \in \mathbb{C}^*$ that $Q\left(\alpha_i + \frac{1}{\zeta}\right) = Q\left(\alpha_i + \frac{1}{\zeta}\right) - Q(\alpha_i) = \frac{1}{\zeta}Q'(\alpha_i) + \frac{1}{\zeta^2}S_i(1/\zeta)$, where $S_i \in \mathbb{C}[z]$ is some polynomial of degree at most $n - 2$ (if $n = 1$, equality doesn't hold). Write similarly $P\left(\alpha_i + \frac{1}{\zeta}\right) - P(\alpha_i) = \frac{1}{\zeta}P'(\alpha_i) +$

$\frac{1}{\zeta^2} R_i(1/\zeta)$, where $\deg R_i = m - 2$. Then

$$\frac{P\left(\alpha_i + \frac{1}{\zeta}\right)}{Q\left(\alpha_i + \frac{1}{\zeta}\right)} = \frac{P(\alpha_i) + \frac{1}{\zeta}P'(\alpha_i) + \frac{1}{\zeta^2}R_i(1/\zeta)}{Q(\alpha_i) + \frac{1}{\zeta}Q'(\alpha_i) + \frac{1}{\zeta^2}S_i(1/\zeta)} = \frac{P(\alpha_i)\zeta^n + P'(\alpha_i)\zeta^{n-1} + R_i(1/\zeta)\zeta^{n-2}}{Q'(\alpha_i)\zeta^{n-1} + S_i(1/\zeta)\zeta^{n-2}},$$

so P/Q has singular part $\frac{P(\alpha_i)}{Q'(\alpha_i)(z-\alpha_i)}$ at α_i . Since $\deg P < \deg Q$, P/Q has no singular part at $z = \infty$. Hence, we write

$$\frac{P(z)}{Q(z)} = \sum_{i \in [n]} \frac{P(\alpha_i)}{Q'(\alpha_i)(z - \alpha_i)} + C$$

for some $C \in \mathbb{C}$. At $z = \infty$, we have $C = 0$, so the result follows.

Given c_i for $i \in [n]$, let $P(z) = \sum_{i \in [n]} \frac{c_i Q(z)}{Q'(\alpha_i)(z - \alpha_i)}$, which has degree at most n . Note that $P(\alpha_i) = c_i$ by L'Hopital's, showing existence. Conversely, if P is such a polynomial, then P has the above formula by the previous result.

4 Power Series

Claim: If $\lim_{n \rightarrow \infty} z_n = A$, then $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i \in [n]} z_i = A$.

Proof: Let z_n be bounded by $M > 0$. Let $\epsilon > 0$. Pick $N \in \mathbb{Z}^+$ such that $n \geq N \rightarrow |z_n - A| \leq \epsilon$. Suppose $n \geq N/\epsilon$. Then

$$\begin{aligned} \left| \frac{1}{n} \left(\sum_{i=1}^N z_i + \sum_{i=N+1}^n z_i \right) - A \right| &= \frac{1}{n} \left| \sum_{i=1}^N (z_i - A) + \sum_{i=N+1}^n (z_i - A) \right| \\ &\leq \frac{1}{n} \sum_{i=1}^N (|z_i| + |A|) + \frac{1}{n} \sum_{i=N+1}^n |z_i - A| \leq \frac{1}{N/\epsilon} N(M + |A|) + \frac{1}{n} (n - N)\epsilon \\ &\leq \epsilon(M + |A| + 1). \end{aligned}$$

The constant $M + |A| + 1$ is independent of ϵ, N , so the result follows.

Claim: The sum of an absolutely convergent series does not change if the terms are rearranged.

Proof: Assume $\sum_{n=1}^{\infty} z_n$ is absolutely convergent. Let $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ be a bijection. Let $\epsilon > 0$. Since $\sum_{n=1}^{\infty} z_n$ is absolutely convergent, pick $N \in \mathbb{Z}^+$ such that for all $n \geq m \geq N$, we have $\sum_{i=m}^n |z_i| \leq \epsilon$. Let $M = \max_{i \in [N]} f^{-1}(i) \geq N$, which exists since the indexing set is finite.

For $n \geq M$, we have

$$\begin{aligned} \left| \sum_{i=1}^n z_i - \sum_{i=1}^n z_{f(i)} \right| &= \left| \left(\sum_{i=1}^N z_i - \sum_{i \in f^{-1}([N])} z_{f(i)} \right) + \left(\sum_{i=N+1}^n z_i - \sum_{i \in [n] - f^{-1}([N])} z_{f(i)} \right) \right| \\ &\leq 2 \sum_{i=N+1}^{\max\{f([n])\}} |z_i| = 2\epsilon, \end{aligned}$$

so $\sum_{i=1}^{\infty} z_i = \sum_{i=1}^{\infty} z_{f(i)}$, as desired.

Discuss the convergence and uniform convergence of the sequence $\{nz^n\}_1^{\infty}$.

For $|z| < 1$, we have that $nz^n \rightarrow 0$. For $|z| \geq 1$, we have $nz^n \rightarrow \infty$. Since the function

$$f(z) = \begin{cases} 0 & \text{if } |z| < 1, \\ \infty & \text{if } |z| \geq 1, \end{cases}$$

is not continuous, $\{nz^n\}_1^{\infty}$ does not converge uniformly.

Claim: Let $|z| < 1$. Then $\lim_{n \rightarrow \infty} nz^n = 0$. It suffices to show if $a \in \mathbb{R}$ such that $0 \leq a < 1$, then $\lim_{n \rightarrow \infty} na^n = 0$. If $a = 0$, we're done, so assume $a > 0$. Let $b = 1/a > 1$. Write $b = 1 + \kappa$ for some $\kappa > 0$. Then for $n \geq 2$, we have

$$\begin{aligned} (1/a)^n &= (1 + \kappa)^n \geq n(n-1)\kappa^2/2 \rightarrow \\ 0 &\leq na^n \leq \frac{2}{\kappa^2(n-1)}. \end{aligned}$$

By Squeeze lemma, since $\lim_{n \rightarrow \infty} \frac{1}{n-1} = 0$, we have $\lim_{n \rightarrow \infty} na^n = 0$, as desired.

Claim: $\{nz^n\}_1^{\infty}$ doesn't converge uniformly on $|z| < 1$.

Proof: The pointwise limit is 0. Let $1 > \epsilon > 0$ and $n \in \mathbb{Z}^+$. We wish to show that there exists $|z| < 1$ such that $|nz^n| \geq \epsilon$. Indeed, $\lim_{z \rightarrow 1} nz^n = n \geq 1$, so there exists $|z| < 1$ such that $|nz^n| > \epsilon$.

Claim: Let (a_n) be a sequence of positive reals. Then

$$\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{a_n} \leq \limsup_{n \rightarrow \infty} \sqrt[n]{a_n} \leq \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}.$$

The middle inequality is automatic. Let $b_n = a_{n+1}/a_n$. Suppose that the set $\{b_n : n \in \mathbb{Z}^+\}$ is unbounded. Then $b_k^+ = \sup\{b_n : n \geq k\} = \infty$, so $\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = \infty$. Then the right inequality is automatic.

Suppose $\{b_n : n \in \mathbb{Z}^+\}$ is bounded, so $\limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} \in \mathbb{R}_{\geq 0}$. Suppose $y > \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$. There exists $N \in \mathbb{Z}^+$ such that $n \geq N \rightarrow b_n^+ < y$. Then $a_{n+1}/a_n \leq b_n^+ < y$ for all $n \geq N$. By Telescoping, $a_n/a_N \leq y^{n-N}$, so $\sqrt[n]{a_n} \leq C^{1/n}y$ for some $C > 0$. It follows that

$$\limsup_{n \rightarrow \infty} \sqrt[n]{a_n} \leq \limsup_{n \rightarrow \infty} C^{1/n}y = y \lim_{n \rightarrow \infty} C^{1/n} = y.$$

Since this holds for arbitrary $y > \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$, we have that

$$\limsup_{n \rightarrow \infty} \sqrt[n]{a_n} \leq \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n},$$

as desired. The other side is similar.

As a corollary, if $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = R$, then $\limsup_{n \rightarrow \infty} \sqrt[n]{a_n} = R$ as well by Squeeze lemma.

5 Analytic Functions as Mappings

Give a precise definition of a single-valued branch of $w = \sqrt{1+z} + \sqrt{1-z}$ in a suitable region, and prove that it is analytic.

Consider the series $f(z) = \sum_{n=0}^{\infty} \binom{\alpha}{n} z^n$. The ratio between coefficients of adjacent terms is

$$\binom{\alpha}{n+1} / \binom{\alpha}{n} = \frac{(\alpha)_{n+1}}{(n+1)!} \cdot \frac{n!}{(\alpha)_n} = \frac{\alpha - n}{n+1} \rightarrow -1$$

as $n \rightarrow \infty$, so it is possible to define $(1+z)^\alpha$ for $|z| < 1$.

Let Ω be $\mathbb{C}$ minus the half-line given by $x \leq 1$ and $y = 0$. Write $w = \sqrt{1+z} + i\sqrt{z-1}$ and take $\sqrt{\cdot}$ to be the same as before. w is analytic as the sum of two analytic functions.

Same with $\log \log z$. We wish for $\log z \leq 0$ to be impossible. Then $0 < z \leq 1$ is subtracted, so Ω is the same as the previous question.

Suppose $f(z) = u + iv$ is analytic in a region Ω and that $F(u, v) = 0$ where F is real-valued with continuous first order derivatives. When can we say that $f(z)$ is constant?

Ex: We have seen $F = v, u^2 + v^2 - C$. Take partials:

$$\begin{cases} \frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \cdot \frac{\partial v}{\partial x} = 0 \\ \frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \cdot \frac{\partial v}{\partial y} = 0 \end{cases} \rightarrow \begin{cases} \frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial x} - \frac{\partial F}{\partial v} \cdot \frac{\partial u}{\partial y} = 0 \\ \frac{\partial F}{\partial v} \cdot \frac{\partial u}{\partial x} + \frac{\partial F}{\partial u} \cdot \frac{\partial u}{\partial y} = 0 \end{cases}$$

The systems does not have a unique solution when $\left(\frac{\partial F}{\partial u}\right)^2 + \left(\frac{\partial F}{\partial v}\right)^2 = 0$, ie $\frac{\partial F}{\partial u} = \frac{\partial F}{\partial v} = 0$. If there do not exist points where ∇F vanishes, then $\nabla u = \nabla v = 0$, from which we conclude f is constant on Ω .

Let C be a circle with center z and radius w such that the set of points of C have the same distance from 0. Then $z = 0$ or $w = 0$.

Proof: Suppose $|z + w| = |z - w|$. Then $z\bar{w} + w\bar{z} = 0$. Replace w with iw . Then $z\bar{i}w + iw\bar{z} = 0 \rightarrow -iz\bar{w} + iw\bar{z} = 0 \rightarrow z\bar{w} = w\bar{z} = 0$. Then $z = 0$ or $w = 0$, as desired.

6 Riemann Sphere

Let $z = x + yi \in \mathbb{C}$. By Similar triangles, the map f from $\mathbb{C} \rightarrow S^2$ is given by $(x, y) \mapsto \left(\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \frac{x^2+y^2-1}{1+x^2+y^2}\right)$. Indeed, as $z \rightarrow \infty$, we have that $\frac{x^2+y^2-1}{1+x^2+y^2} \rightarrow 1$. Note that indeed

$$\left(\frac{2x}{1+x^2+y^2}\right)^2 + \left(\frac{2y}{1+x^2+y^2}\right)^2 + \left(\frac{x^2+y^2-1}{1+x^2+y^2}\right)^2 = 1$$

so as $z \rightarrow \infty$, $f(z) \rightarrow (0, 0, 1)$. If $z, w \in \mathbb{C}$, then the associated distance in S^2 is (by Similar triangles)

$$d(z, w) = \frac{2|z - w|}{\sqrt{(1 + |z|^2)(1 + |w|^2)}}.$$

7 Linear Transformations

Let $w = \frac{az+b}{cz+d}$ be a linear transformation such that $\mathbb{R}$ is sent to $\mathbb{R}$. Then w can be rewritten with $a, b, c, d \in \mathbb{R}$.

Proof: Suppose $c = 0$, so $w = \lambda z + \mu$ for $\lambda, \mu \in \mathbb{C}$. $w(0) = \mu \in \mathbb{R}$ and $w(1) = \lambda + \mu \in \mathbb{R}$, so $\mu \in \mathbb{R}$ as well. Suppose $d = 0$. Then $w = \frac{\lambda z + \mu}{z}$ for $\lambda, \mu \in \mathbb{C}$. Then $\lambda r + \mu \in \mathbb{R}$ for all $r \in \mathbb{R}^\times$. Take $r = 1, -1$, so $\lambda, \mu \in \mathbb{R}$. Assume $c, d \neq 0$.

$\mathbb{R}$ is a closed subset of $\mathbb{C}$, so $\lim_{r \rightarrow \infty} w(r) = a/c \in \mathbb{R}$. $w(0) = b/d \in \mathbb{R}$ as well. Write $a = \lambda c$ and $b = \mu d$ for $\lambda, \mu \in \mathbb{R}$. Then $w = \frac{\lambda cz + \mu d}{cz + d}$. Assume $c \neq d$. Then $-\lambda c + \mu d = r(-c + d) \rightarrow (r - \lambda)c = (r - \mu)d$. Since $c, d \neq 0$ either $r = \lambda = \mu$ or $\frac{c}{d} = \frac{r - \mu}{r - \lambda} \in \mathbb{R}$.

If $\lambda = \mu$, then $w = \lambda$, absurd. Else, write

$$w = \frac{(a/c) \cdot (c/d)z + (b/d)}{(c/d)z + 1},$$

which has real coefficients.

Finally, if $c = d$, then $w = \frac{\lambda z + \mu}{z + 1}$, which also has real coefficients.

Suppose $z_2, z_3, z_4 \in \mathbb{C}^*$ are distinct. Suppose $z_2, z_3, z_4 \neq \infty$. The linear transformation

$$S = \frac{z - z_3}{z - z_4} \cdot \frac{z_2 - z_4}{z_2 - z_3}$$

If z_2, z_3, z_4 is ∞ , S becomes

$$\frac{z - z_3}{z - z_4}, \frac{z_2 - z_4}{z - z_4}, \frac{z - z_3}{z_2 - z_3}.$$

Claim: Let S be a linear transformation fixing $1, 0, \infty$. Then $S = z$.

Proof: Let $S = \frac{az+b}{cz+d}$. Note $c = 0$, else $S(\infty) = a/c \in \mathbb{C}$. Write $S = \lambda z + \mu$. Then $\mu = 0$ and $\lambda = 1$, ie $S = z$.

If $z_1, z_2, z_3, z_4 \in \mathbb{C}^*$ are distinct and T is a linear transformation, then $(Tz_1, Tz_2, Tz_3, Tz_4) = (z_1, z_2, z_3, z_4)$.

Proof: Let S be the linear transformation sending (z_2, z_3, z_4) to $(1, 0, \infty)$. Let R be the linear transformation sending (Tz_2, Tz_3, Tz_4) to $(1, 0, \infty)$. Then $R = ST^{-1}$. In particular,

$$(Tz_1, Tz_2, Tz_3, Tz_4) = R(Tz_1) = S(z_1) = (z_1, z_2, z_3, z_4),$$

as desired.

$(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_4} : \frac{z_2 - z_3}{z_2 - z_4}$. Then $(z_2, z_1, z_3, z_4) = \frac{z_2 - z_3}{z_2 - z_4} : \frac{z_1 - z_3}{z_1 - z_4} = 1/\lambda$.
 $(z_1, z_2, z_4, z_3) = \frac{z_1 - z_4}{z_1 - z_3} : \frac{z_2 - z_4}{z_2 - z_3} = 1/\lambda$. $(z_3, z_2, z_1, z_4) = \frac{z_3 - z_1}{z_3 - z_4} : \frac{z_2 - z_1}{z_2 - z_4} = \mu$. We have

$$\begin{aligned} \frac{1}{\lambda} + \frac{1}{\mu} &= \frac{(z_1 - z_4)(z_2 - z_3)}{(z_1 - z_3)(z_2 - z_4)} + \frac{(z_3 - z_4)(z_2 - z_1)}{(z_3 - z_1)(z_2 - z_4)} \\ &= \frac{z_1 z_2 - z_2 z_4 - z_1 z_3 + z_3 z_4 - z_2 z_3 + z_2 z_4 + z_1 z_3 - z_1 z_4}{(z_1 - z_3)(z_2 - z_4)} \\ &= \frac{z_1 z_2 + z_3 z_4 - z_2 z_3 - z_1 z_4}{(z_1 - z_3)(z_2 - z_4)} = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_3)(z_2 - z_4)} = 1. \end{aligned}$$

Hence, $\mu = \frac{\lambda}{\lambda - 1}$.

Let $T : \mathbb{C}^* \rightarrow \mathbb{C}^*$ be a linear transformation and denote by C the image of $\mathbb{R}$ under T . Let $w, w^* \in \mathbb{C}^*$. w, w^* are said to be symmetric with respect to C and T if $w = Tz$ and $w^* = T\bar{z}$.

Claim: If w, w^* are symmetric with respect to C, T , then w, w^* are symmetric to C, S for any $S : \mathbb{R} \rightarrow C$. Proof: Suppose $w = Tz$ and $w^* = T\bar{z}$. Then $w = S(S^{-1}Tz)$ and $w^* = S(S^{-1}T\bar{z})$. The linear transformation $S^{-1}T$ sends $\mathbb{R}$ to itself, and is a real transformation. Thus, $S^{-1}T\bar{z} = \overline{S^{-1}Tz}$. Let $z' = S^{-1}Tz$, so $w = Sz'$ and $w^* = S\bar{z}'$.

Def: The points z, z^* are said to be symmetric with respect to the circle C through z_1, z_2, z_3 iff $(z^*, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)}$. Let $S(z) = (z, z_1, z_2, z_3)$. We have $S(z^*) = \overline{S(z)}$. $S^{-1} : \mathbb{R} \rightarrow C$ is our transformation. We have $z^* = S^{-1}(\overline{S(z)})$.

Suppose $(z, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)}$. Then $(z, z_1, z_2, z_3) \in \mathbb{R}$, so $z \in C$.

Suppose $z_3 = \infty$. The condition for symmetry becomes $\frac{z^* - z_2}{z^* - z_3} \cdot \frac{z_1 - z_3}{z_1 - z_2} = \overline{\frac{z - z_2}{z - z_3} \cdot \frac{z_1 - z_3}{z_1 - z_2}} \rightarrow \frac{z^* - z_2}{z_1 - z_2} = \overline{\frac{z - z_2}{z_1 - z_2}}$, so z^* is the reflection of z across the line joining z_1, z_2 .

Claim: $(1/z_1, 1/z_2, 1/z_3, 1/z_4) = (z_1, z_2, z_3, z_4)$. Proof:

$$\frac{1/z_1 - 1/z_3}{1/z_1 - 1/z_4} : \frac{1/z_2 - 1/z_3}{1/z_2 - 1/z_4} = \frac{z_3 - z_1}{z_4 - z_1} : \frac{z_3 - z_2}{z_4 - z_2} = (z_1, z_2, z_3, z_4).$$

Suppose C is a finite circle with center a , radius R . Then

$$\begin{aligned} \overline{(z, z_1, z_2, z_3)} &= \overline{(z - a, z_1 - a, z_2 - a, z_3 - a)} \\ &= \left(\bar{z} - \bar{a}, \frac{R^2}{z_1 - a}, \frac{R^2}{z_2 - a}, \frac{R^2}{z_3 - a} \right) \\ &= \left(\frac{\bar{z} - \bar{a}}{R^2}, \frac{1}{z_1 - a}, \frac{1}{z_2 - a}, \frac{1}{z_3 - a} \right) \\ &= \left(\frac{R^2}{\bar{z} - \bar{a}} + a, z_1, z_2, z_3 \right), \end{aligned}$$

so $z^* = R^2/(\bar{z} - \bar{a}) + a$, or $(z^* - a)(\bar{z} - \bar{a}) = R^2$.

If a linear transformation carries C_1 to a circle C_2 , then it transforms a pair of symmetric points with respect to C_1 to a pair of symmetric points with respect to C_2 .

Proof: Let $z_1, z_2, z_3 \in C_1$ be distinct. Let T be our transformation and $w_1, w_2, w_3 \in C_2$ the corresponding points under T . Let S_1, S_2 be linear transformations sending $1, 0, \infty$ to $(z_1, z_2, z_3), (w_1, w_2, w_3)$. By uniqueness, we have $T = S_2 S_1^{-1}$. Suppose $z = S_1 z_0$ and $z^* = S_1 \bar{z}_0$. Then $Tz = S_2 z_0$ and $Tz^* = S_2 \bar{z}_0$, ie Tz, Tz^* are symmetric with respect to C_2 .

Let $C \in \mathbb{C}^*$. Fix $z_1, z_2, z_3 \in C$, so $C : (z, z_1, z_2, z_3) \in \mathbb{R}$. Note $(z, z_1, z_2, z_3) = (1/z, 1/z_1, 1/z_2, 1/z_3)$, so C' is the circle through $1/z_1, 1/z_2, 1/z_3$.

(1). Reflect the imaginary axis in the circle $|z - 2| = 1$. Let $z = iy$, where $y \in \mathbb{R}$. Then $(\bar{z} - 2)(z^* - 2) = 1 \rightarrow z^* = 2 - \frac{1}{2+iy} = 2 - \frac{2-iy}{4+y^2} = \frac{6+2y^2+iy}{4+y^2}$. Let $z = t(1+i)$. Then

$$\begin{aligned} z^* &= 2 + \frac{1}{t(1-i) - 2} = 2 + \frac{1}{(t-2) - ti} \\ &= 2 + \frac{(t-2) + ti}{2t^2 - 4t + 4} = \frac{4t^2 - 7t + 6 + ti}{2(t^2 - 2t + 2)}. \end{aligned}$$

Let $z = \cos \theta + i \sin \theta$. Then

$$\begin{aligned} z^* &= 2 + \frac{1}{(\cos \theta - 2) - i \sin \theta} = 2 + \frac{(\cos \theta - 2) + i \sin \theta}{(\cos \theta - 2)^2 + \sin^2 \theta} \\ &= 2 + \frac{(\cos \theta - 2) + i \sin \theta}{5 - 4 \cos \theta} = \frac{8 - 7 \cos \theta + i \sin \theta}{5 - 4 \cos \theta}. \end{aligned}$$

Find the most general linear transformation of the circle $|z| = R$ into itself.

Let $S = \frac{az+b}{cz+d}$ such that $|S(z)| = R$ for all $z \in \mathbb{R}$. Assume $R = 1$. We have $|az + b| = |cz + d| \rightarrow a\bar{a}x^2 + (a\bar{b} + b\bar{a})x + b\bar{b} = c\bar{c}x^2 + (c\bar{d} + d\bar{c})x + d\bar{d}$, so

$$(|a|^2 - |c|^2)x^2 + (a\bar{b} + b\bar{a} - c\bar{d} - d\bar{c})x + (|b|^2 - |d|^2) = 0$$

holds for all $x \in \mathbb{R}$, so equivalently, $|a| = |c|$, $|b| = |d|$, and $a\bar{b} + b\bar{a} = c\bar{d} + d\bar{c}$. Let $c = au, d = bv$ for $|u| = |v| = 1$. Then

$$\begin{aligned} a\bar{b} + b\bar{a} &= a\bar{b}(u/v) + b\bar{a}(v/u) \rightarrow a\bar{b}(1 - u/v) = b\bar{a}(v/u - 1) \\ \rightarrow (v - u)(a\bar{b}/v - b\bar{a}/u) &= 0 \rightarrow v = u \text{ or } a\bar{b}/v = b\bar{a}/u \end{aligned}$$

Note $a, b, c, d \neq 0$ (because S degenerates), so $v = \frac{a\bar{b}}{b\bar{a}}u$. We may scale the coefficients so $a \in \mathbb{R}$. Then $v = \frac{\bar{b}}{b}u$. Then $d = vb = \frac{\bar{b}}{b}ub = \bar{b}u$. In the first case, S becomes $\frac{az+b}{u(az+b)} = 1/u$, absurd. In the second, $S = \frac{az+b}{u(az+b)} = \frac{z+b/a}{u(z+b/a)}$, so the general form of S is

$$S = \lambda \frac{z - \mu}{z - \bar{\mu}},$$

where $|\lambda| = R$, which we can check to work. Composing $S_2 S_1^{-1}$ gives the desired linear transformation from $|z| = R$ to itself.

Suppose S is a linear transformation of $|z| = R$ to itself. Assume $S^{-1}(0) = a$, so $S^{-1}(\infty) = R^2/\bar{a}$. Let $S^{-1}(R) = c$ with $|c| = R$. Then have $(S(z), R, 0, \infty) =$

$(z, c, a, R^2/\bar{a})$, so

$$\frac{S(z)}{S(z) - \infty} : \frac{R}{R - \infty} = \frac{z - a}{z - R^2/\bar{a}} : \frac{c - a}{c - R^2/\bar{a}}$$

$$S(z) = R \frac{z - a}{\bar{a}z - R^2} \cdot \frac{\bar{a}c - R^2}{c - a}$$

Compute

$$\left| \frac{\bar{a}c - R^2}{c - a} \right| = \left| \frac{R^2\bar{a}/\bar{c} - R^2}{R^2/\bar{c} - a} \right| = R^2 \left| \frac{a - c}{R^2 - \bar{a}c} \right| \rightarrow \left| \frac{\bar{a}c - R^2}{c - a} \right| = R.$$

Write $S(z) = \lambda \frac{z - a}{\bar{a}z - R^2}$ with $|\lambda| = R^2$.

$$S'(z) = \frac{a(cz+d) - (az+b)c}{(cz+d)^2} = \frac{ad-bc}{(cz+d)^2} \neq 0, w = S(z) \text{ is conformal for } z \neq -d/c, \infty.$$

Let $z_1, z_2, z_3 \in \mathbb{C}$. (z_1, z_2, z_3) is said to be positively oriented if $\Im(\infty, z_1, z_2, z_3) > 0$. Then $\Im(z_1, \infty, z_2, z_3) < 0$. Let $z = a + bi$. Then

$$\begin{aligned} \frac{z}{z-1} &= \frac{a+bi}{(a-1)+bi} = \frac{(a+bi)(a-1-bi)}{(a-1)^2+b^2} \\ &= \frac{a(a-1)+b^2-bi}{(a-1)^2+b^2}, \end{aligned}$$

so $\Im z, \Im \frac{z}{z-1}$ have opposite sign, so $\Im(z_2, \infty, z_1, z_3) > 0 \rightarrow \Im(\infty, z_2, z_1, z_3) < 0$, $\Im(\infty, z_2, z_3, z_1) > 0$. $\Im(\infty, z_1, z_3, z_2) < 0$. $\Im(\infty, z_1, z_3, z_2) < 0 \rightarrow \Im(z_1, \infty, z_3, z_2) > 0 \rightarrow \Im(z_3, \infty, z_1, z_2) < 0 \rightarrow \Im(\infty, z_3, z_1, z_2) > 0$, $\Im(\infty, z_3, z_2, z_1) < 0$. In all, the positively oriented triples are

$$(z_1, z_2, z_3), (z_2, z_3, z_1), (z_3, z_1, z_2)$$

and negatively oriented are

$$(z_1, z_3, z_2), (z_2, z_1, z_3), (z_3, z_2, z_1)$$

Suppose $z_1, z_2, z_3, z'_1, z'_2, z'_3 \in C$ are positively oriented. Then $\Im(w, z_1, z_2, z_3), \Im(w, z'_1, z'_2, z'_3)$ have the same sign for all $w \in \mathbb{C}^*$. We have

$$\Im \frac{\infty - z_2}{\infty - z_3} : \frac{z_1 - z_2}{z_1 - z_3} = \Im \frac{z_1 - z_3}{z_1 - z_2} > 0.$$

Let C be a circle, S be LT sending $(1, 0, \infty)$ to (z_1, z_2, z_3) .

Let $(z_1, z_3, z_4), (z_2, z_3, z_4)$ be positively oriented. Show $(z_1, z_2, z_3, z_4) > 0$. We have

$$\Im \frac{z_1 - z_4}{z_1 - z_3}, \Im \frac{z_2 - z_4}{z_2 - z_3} > 0$$

So we have $z_1 = a_1 + b_1i, z_2 = a_2 + b_2i$ such that b_1, b_2 have the same sign and $z_1/z_2 \in \mathbb{R}$. Show that $z_1/z_2 > 0$. Write

$$\frac{z_1}{z_2} = \frac{a_1 + b_1i}{a_2 + b_2i} = \frac{(a_1 + b_1i)(a_2 - b_2i)}{a_2^2 + b_2^2} = \frac{(a_1a_2 + b_1b_2) + (b_1a_2 - a_1b_2)i}{a_2^2 + b_2^2} \rightarrow b_1a_2 = a_1b_2,$$

so a_1, a_2 have the same sign. Then $a_1a_2 + b_1b_2 > 0 + 0 = 0$ (a_1, a_2 could both be zero). If b_1, b_2 have distinct signs, so do a_1, a_2 .

Let $(a + R, a + iR, a - R)$ represent C . The triple has positive orientation since

$$\frac{a + R - (a - R)}{a + R - (a + iR)} = \frac{2R}{(1 - i)R} = 1 + i$$

has positive imaginary part. Suppose $z \in \mathbb{C}$ such that $(z, a + R, a + iR, a - R) < 0$. Let $z = a + Rw$. We have

$$\begin{aligned} & \frac{(a + wR) - (a + iR)}{(a + wR) - (a - R)} : \frac{(a + R) - (a + iR)}{(a + R) - (a - R)} = \frac{w - i}{w + 1} : \frac{1 - i}{2} \\ & = \frac{(w - i)(1 + i)}{w + 1} = \frac{w + 1 - i + iw}{w + 1} = 1 + i \frac{w - 1}{w + 1} \end{aligned}$$

Let $w = a + bi$. Then

$$\begin{aligned} \frac{w - 1}{w + 1} &= \frac{(a - 1) + bi}{(a + 1) + bi} = \frac{((a - 1) + bi)((a + 1) - bi)}{(a + 1)^2 + b^2} \\ &= \frac{(a^2 - 1 + b^2) + ci}{(a + 1)^2 + b^2}, \end{aligned}$$

so we wish for $(a^2 + b^2 - 1) < 0 \rightarrow |w| < 1$, or equivalently $|z - a| < R$.

Consider $C = \mathbb{R}$. Write $S = \frac{az+b}{cz+d}$ for $a, b, c, d \in \mathbb{R}$. Let $z = x + yi$. Then

$$\begin{aligned} S(z) &= \frac{a(x + yi) + b}{c(x + yi) + d} = \frac{(ax + b) + ayi}{(cx + d) + cyi} \\ &= \frac{((ax + b) + ayi)((cx + d) - cyi)}{(cx + d)^2 + (cy)^2} \\ &= \frac{(ax + b)(cx + d) + acy^2 + (ay(cx + d) - (ax + b)cy)i}{(cx + d)^2 + (cy)^2} \\ \rightarrow \Im S(z) &= \frac{ay(cx + d) - (ax + b)cy}{(cx + d)^2 + (cy)^2} = \frac{(ad - bc)y}{(cx + d)^2 + (cy)^2} = \frac{(ad - bc)}{|cz + d|^2} \Im z \end{aligned}$$

$$cz + d = c(x + yi) + d = (cx + d) + cyi.$$

Suppose $z = \frac{az+b}{cz+d} \rightarrow cz^2 + dz = az + b \rightarrow cz^2 + (d - a)z - b = 0$, so

$$(d-a)^2 + 4bc = 0.$$

Suppose S carries a, b to a', b' . Then S can be written as

$$\frac{w-a'}{w-b'} = k \frac{z-a}{z-b}$$

for some $k \in \mathbb{C}$.

Proof: Let S also send c to c' . Consider the equation in $w, z \in \mathbb{C}$: $(w, c', a', b') = (z, c, a, b)$. The LHS is a linear function of w and the RHS is a linear function of z , so w is a linear function of z . Given the common value u of both sides, there are unique z, w giving that value. When $u = 0$, we have $w = a', z = a$. When $u = \infty$, we have $w = b', z = b$. Then $u = 1$, we have $w = c', z = c$. Hence, w sends a, b, c to a', b', c' . By definition of Cross product, the result follows.

Suppose for $a \neq b$ and $|k| = 1$ that $\frac{w-a}{w-b} = k \cdot \frac{z-a}{z-b}$. Let $S = w(z)$. I claim that S^n is elliptic for all $n \geq \mathbb{Z}^+$. Note S^n fixes a, b . Write

$$\begin{aligned} (w-a)(z-b) &= k(w-b)(z-a) \\ wz - az - bw + ab &= kwz - kbz - kaw + kab \\ w(z-b-kz+ka) &= -kbz + kab + az - ab \\ w &= \frac{(a-kb)z + (k-1)ab}{(1-k)z + (ka-b)} \end{aligned}$$

Compute

$$\begin{aligned} \det w &= -(ka-b)(kb-a) + (k-1)^2 ab \\ &= -k^2 ab + ka^2 + kb^2 - ab + k^2 ab - 2kab + ab \\ &= k(a-b)^2 \end{aligned}$$

$$\text{and } \text{tr } w = (k+1)(a-b).$$

Let w' be the scaled version of w . Then $\text{tr } w' = \text{tr } w / \sqrt{\det w} = \frac{(k+1)(a-b)}{(a-b)\sqrt{k}} = \frac{k+1}{\sqrt{k}}$. Write $k = \cos \theta + i \sin \theta$. Then

$$\frac{k+1}{\sqrt{k}} = \frac{(\cos \theta + 1) + i \sin \theta}{\cos \theta/2 + i \sin \theta/2} = \frac{2 \cos^2 \theta/2 + 2i \cos \theta/2 \sin \theta/2}{\cos \theta/2 + i \sin \theta/2} = 2 \cos \theta/2,$$

Note if $k = 1$, then $\frac{w-a}{w-b} = \frac{z-a}{z-b}$. The map $z \mapsto \frac{z-a}{z-b}$ is invertible, so $w = z$. Else, $\sqrt{k} \neq \pm 1$, so $\left| \frac{k+1}{\sqrt{k}} \right| < 2$.

Suppose $k > 0$ is not equal to 1. Then by AM-GM, $\frac{1+k}{\sqrt{k}} > 2$.

Suppose $S^n z = z$ for some $n \in \mathbb{Z}^+$ (we assume $n \neq 0$ and if $n < 0$, we can invert).

Let ζ be a primitive n th root of unity. We have that $[S] \sim \begin{bmatrix} \zeta^{k_1} & 0 \\ 0 & \zeta^{k_2} \end{bmatrix}$ for some $k_1, k_2 \in \mathbb{Z}$. Then in this representation, $\det S = \zeta^{k_1+k_2}$ and $\text{tr } S = \zeta^{k_1} + \zeta^{k_2}$, so

$$\begin{aligned} \text{sgn } S &= \frac{\zeta^{k_1} + \zeta^{k_2}}{\zeta^{(k_1+k_2)/2}} \in \mathbb{R} \text{ (by considering conjugate)} \\ |\text{sgn } S| &\leq |\zeta^{(k_1-k_2)/2}| + |\zeta^{(k_2-k_1)/2}| = 2, \end{aligned}$$

so S is elliptic.

Suppose $z = \frac{\alpha z + \beta}{\gamma z + \delta}$ has one fixed point (and $\gamma \neq 0$). Then $\gamma z^2 + \delta z = \alpha z + \beta \rightarrow \gamma z^2 + (\delta - \alpha)z - \beta = 0 \rightarrow (\delta - \alpha)^2 + 4\gamma\beta = 0 \rightarrow (\delta + \alpha)^2 + 4(\gamma\beta - \delta\alpha) = 0 \rightarrow \text{tr }^2 w = 4 \det w \rightarrow \text{sgn } w = \pm 2$.

8 Level Curves

Let $w = z^2$, so $u = x^2 - y^2, v = 2xy$. Suppose (x, y) is a common point of $u = u_0$ and $v = v_0$. Then the curves are orthogonal at (x, y) .

Proof: We have $\nabla u = \langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \rangle$ and $\nabla v = \langle \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \rangle = \langle -\frac{\partial u}{\partial y}, \frac{\partial u}{\partial x} \rangle$, so $\nabla u, \nabla v$ are perpendicular.

Let $\Omega_1 = \mathbb{R} \times (-\frac{\pi}{2}, \frac{\pi}{2})$, $\Omega_2 = \mathbb{R}^+ \times \mathbb{R}$, $\Omega_3 = (E^2)^\circ$. I claim that we can map conformally from Ω_1 to Ω_3 .

Proof: Let $f_1 : \Omega_1 \rightarrow \Omega_2$ be the map $f_1(z) = e^z$. Let $z = a + bi$ with $|b| < \pi/2$ and $a \in \mathbb{R}$. Then $f_1(z) = e^a(\cos b + i \sin b)$. $\cos b + i \sin b$ achieves any number on the unit circle with positive real part. e^a achieves any modulus. It is easy to see that f_1 is bijective.

Let $f_2 : \Omega_2 \rightarrow \Omega_3$ be the map $f_2(z) = \frac{z-1}{z+1}$. Then $f_2 f_1$ is our desired map.

Map the region between $|z| = 1$ and $|z - 1/2| = 1/2$ on a half plane. Map $z \mapsto z - 1$ so we have two circles tangent at the origin. Map $z \mapsto 1/z$ so our region is the space between the lines $x = -1, -1/2$. Map $z \mapsto 2\pi i(z + 3/4)$, so our region is the space between $y = -\pi/2, \pi/2$. Then map $z \mapsto e^z$, so we are left with the half plane $x > 0$. Together, the map is

$$z \mapsto \exp \left(2\pi i \left(\frac{1}{z-1} + \frac{3}{4} \right) \right).$$

9 Riemann Surfaces

Suppose $z + \frac{1}{z} = w + \frac{1}{w}$. Then $(z - w) - \frac{z-w}{wz} = 0 \rightarrow z = w$ or $z = 1/w$.

Review of Abel's theorem. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ have radius of convergence $R > 0$ such that $\sum_{n=0}^{\infty} a_n z_0^n$ converges with $|z_0| = R$. Then $f(z) \rightarrow f(z_0)$ if $|z_0 - z|/(R - |z|)$ is bounded and $z \rightarrow z_0$.

Proof: Let $b_n = a_n z_0^n$. Let $g(z) = \sum_{n=0}^{\infty} b_n z^n$ has radius of convergence 1. Indeed,

$$\limsup_{n \rightarrow \infty} \sqrt[n]{b_n} = \limsup_{n \rightarrow \infty} |z_0| \sqrt[n]{a_n} = |z_0| \limsup_{n \rightarrow \infty} \sqrt[n]{a_n} = R \cdot \frac{1}{R} = 1.$$

Let $z \rightarrow z_0$ such that $|z_0 - z|/(R - |z|)$ is bounded. Let $z = z_0 w$. Then $w \rightarrow 1$ such that $|1 - w|/(1 - |w|)$ is bounded. We wish to show $g(w) \rightarrow g(1)$ (we know $g(1) = f(z_0)$ is convergent).

Let $B = \sum_{n=0}^{\infty} b_n$ and $s_n = \sum_{k=0}^n b_k$. Write

$$\begin{aligned} g(z) &= \lim_{n \rightarrow \infty} \left(b_0 + \sum_{k=1}^n (s_k - s_{k-1}) z^k \right) \\ &= \lim_{n \rightarrow \infty} (b_0 + (s_1 - s_0)z + (s_2 - s_1)z^2 + \cdots + (s_n - s_{n-1})z^n) \\ &= \lim_{n \rightarrow \infty} b_0 - s_0 z + (1 - z)(s_1 z + s_2 z^2 + \cdots + s_{n-1} z^{n-1}) + s_n z^n \\ &= \lim_{n \rightarrow \infty} (1 - z)(s_0 + s_1 z + s_2 z^2 + \cdots + s_{n-1} z^{n-1}) + s_n z^n \\ &= (1 - z) \sum_{n=0}^{\infty} s_n z^n + B. \end{aligned}$$

Let $\epsilon > 0$. Let s_n be bounded by M . Pick $N \in \mathbb{Z}^+$ such that $|\sum_{k=n}^{\infty} s_k z^k| < \epsilon$ for $n \geq N$. We bound

$$\begin{aligned} |g(z) - B| &= |1 - z| \cdot \left| \sum_{n=0}^{\infty} s_n z^n \right| \\ &= |1 - z| \cdot \left(\left| \sum_{n=0}^{N-1} s_n z^n \right| + \sum_{n=N}^{\infty} M |z|^n \right) \\ &\leq |1 - z| \cdot \left(C + \frac{M}{1 - |z|} \right) \end{aligned}$$